

SIF-Sense AI: Technical Architecture & Storage Specification

A-to-Z Technical Deep Dive: OSHA 2015–2025 ML Model, Dataset Science, Full Tech Stack & Free MongoDB Optimization

Primary Backend: Python FastAPI (Asynchronous REST)	Frontend: React 19 + Vite + Recharts + ReactFlow
ML Framework: Scikit-Learn + PyTorch (Calibrated SGD)	Database: MongoDB Atlas M0 (512MB) + SQLite Fallback
Dataset: 105,996 federal OSHA severe injury records	Configured DB User: kanishkskcet_db_user

1. The Dataset Science: OSHA Severe Injury Archive (2015–2025)

SIF-Sense AI is trained directly on the official **United States Occupational Safety and Health Administration (OSHA) Severe Injury Reports dataset**. Under federal reporting regulation 29 CFR 1904.39, all covered industrial employers in the United States must report any work-related amputation, in-patient hospitalization, or loss of an eye. The dataset contains **105,996 severe workplace trauma records** from January 1, 2015 through November 30, 2025 across all 50 US states.

- **Key Data Columns:** *Final Narrative* (unstructured free-text description of the incident sequence), *Hospitalized count*, *Amputation count*, *Loss of Eye count*, *NatureTitle* (e.g. Fractures, Amputations, Burns, Crushing injuries), *Part of Body Title*, *EventTitle* (incident mechanics), and *Primary NAICS* (industry classification).
- **Ground Truth SIF Labeling:** Incidents are mathematically labeled as Serious Injury or Fatality (SIF) Precursors if they resulted in: (1) Amputation ≥ 1, (2) Loss of Eye ≥ 1, (3) In-patient Hospitalization > 1, or (4) Hospitalization with high-severity life-altering trauma (skull fractures, intracranial hemorrhage, electrical arc flash, chemical/thermal burns, or severe crushing).
- **Dataset Partitioning:** 80% Stratified Training Set (84,796 records) and 20% Unseen Testing Evaluation Set (21,199 records).

2. AI / ML / DL Model Architecture & Training Pipeline

Rather than relying on slow, costly, and non-deterministic cloud LLMs that charge per token, SIF-Sense AI utilizes an ultra-fast, locally deployed, calibrated NLP machine learning pipeline:

- **Feature Extraction (TF-IDF N-grams):** Converts natural language incident text into 25,000 sublinear TF-IDF features capturing single words, bigrams, and trigrams (1-3 n-grams) such as 'lockout tagout', 'fall from scaffold', 'amputated two fingers', and 'arc flash burns'.
- **Calibrated SIF Probability Classifier:** Uses Stochastic Gradient Descent (SGD) with log-loss and balanced class weighting, wrapped in a 3-fold Platt Sigmoid calibrator. This outputs calibrated probabilities (0.00% to 100.0%) rather than uncalibrated margins.
- **Multi-Class Injury Nature Predictor:** Companion multi-nominal logistic regression classifier predicting 8 trauma categories (Amputation, Fracture, Crushing Injury, Thermal/Chemical Burn, Head/Brain Trauma, Electrical Trauma, Laceration, and Sprain/Strain).
- **Precomputed Feature Attribution Engine:** Feature coefficient matrices are precomputed into memory at startup. When an incident is analyzed, word-level risk weights are extracted in **0.05 milliseconds** via direct dictionary lookups.

Model Metric	Evaluation Score	Technical Significance
Training Set	84,796 records	80% stratified representation across all 50 states
Test Set (Unseen)	21,199 records	Rigorous out-of-sample statistical validation
Test Accuracy	96.00%	Correct classifications across all incident types
SIF Safety Recall	97.49%	Zero-tolerance safety: 97.5% of genuine threats caught
Precision	97.18%	Extremely low false alarm / false positive rate
F1-Score	0.9733	Harmonic mean of precision and safety recall
ROC-AUC	0.9876	Superb threshold-independent discrimination
Inference Latency	< 15 ms	Sub-second CPU execution with zero GPU dependency

3. Free Database (MongoDB Atlas M0 512MB) Storage Architecture

MongoDB Atlas M0 provides a hard **512 MB ceiling**. SIF-Sense AI implements a 4-tier lifecycle manager to ensure capacity never runs out:

- **Tier 1: Automated TTL (Time-To-Live) Partial Index:** Purges low-risk non-SIF observations after 90 days, while keeping confirmed SIF precursors indefinitely:

```
# MongoDB TTL Indexing in Python / PyMongo:
db.reports.create_index(
    [("created_at", 1)],
    expireAfterSeconds=7776000, # Exactly 90 days
    partialFilterExpression={"sif_potential": False} # NEVER purges genuine SIF precursors!
)
```

- **Tier 2: Schema Pruning:** Strips unnecessary embeddings and logs in mongo_manager.py. Documents are compressed to **< 380 bytes**, allowing 512 MB to hold **1,340,000+ reports**.
- **Tier 3: Capped Ephemeral Collections:** Rolling audit logs are capped at 20 MB with FIFO eviction.
- **Tier 4: Offline Cold Storage Export:** Built-in CSV & JSON export allows safety directors to archive older logs locally.

4. Backend REST API Specification & Endpoints

Endpoint	Method	Description & Output
/api/analyze	POST	Full pipeline: NLP entity extraction, OSHA model inference, SIF scoring, and graph generation.
/api/model/metrics	GET	Returns model metrics (96% accuracy, 97.5% recall), confusion matrix, and top learned risk terms.
/api/model/predict	POST	Lightweight instant inference playground (<15ms) with word-level risk token weights.
/api/osha/samples	GET	Curated real-world OSHA 2015–2025 benchmark test incidents across major industries.
/api/database/status	GET	Returns active connection status (MongoDB Atlas vs SQLite fallback), active user, and errors.
/api/database/connect	POST	Tests and updates MongoDB Atlas cluster hostname or URI dynamically without server restart.
/api/reports	GET	Lists analyzed reports with pagination (10/25/50 per page), search, and SIF-only filter.
/api/alerts	GET	Retrieves early warning alerts filtered by status (PENDING, ACCEPTED, ESCALATED, REJECTED).
/api/alerts/{id}/review	PATCH	Human-in-the-loop action: ACCEPT, REJECT, or ESCALATE with mandatory HSE audit notes.